

Machine Learning for Type-II Diabetes Screening in India

Benchmarking Classic Models on NFHS-5 Data with Imbalance Correction and Explainable AI

Dataset	NFHS-5 Individual Recode, 2019–21 135,939 women 36 states/UTs
Target	Self-reported diagnosis of Type-II diabetes Prevalence: 0.90%
Models	Logistic Regression · SVM (RBF) · Random Forest · Naive Bayes · AdaBoost · XGBoost
Strategies	None (baseline) · SMOTE oversampling · Class-weighted learning
Explainability	SHAP TreeExplainer (RF + XGBoost) · Odds Ratios (Logistic Regression)

ABSTRACT

Background: Type-II diabetes burden in India continues to rise, yet population-level screening remains resource-intensive. The National Family Health Survey-5 (NFHS-5) provides a nationally representative dataset of socio-demographic and health indicators for women aged 15–49, offering an opportunity to evaluate machine learning for scalable risk prediction.

Objective: To benchmark six classic machine learning models for Type-II diabetes prediction on NFHS-5 data, comparing three class-imbalance correction strategies and integrating SHAP-based explainability.

Methods: Self-reported diabetes labels were extracted from NFHS-5 individual microdata for 135,939 women across 36 states (~0.90% prevalence). Six models were evaluated under three imbalance strategies with GridSearchCV tuning (5-fold stratified CV, ROC-AUC scoring) on a state-stratified 80/20 split. SHAP TreeExplainer was applied to tree-based models; odds ratios derived from Logistic Regression coefficients.

Results: SVM (RBF) with class-weighted learning achieved the highest ROC-AUC (0.745) and recall (0.616). SHAP analysis identified Hypertension, Age, and BMI as the top three predictors. Tobacco and Alcohol showed artifactual inverse associations due to low prevalence and self-reporting bias.

Conclusion: Classic ML models can support Type-II diabetes risk stratification using existing NFHS-5 variables. SMOTE and class-weighted learning improve minority-class recall but involve precision-recall tradeoffs. SHAP-derived importance confirms clinically meaningful predictors and supports policy interpretability.

Keywords: Type-II diabetes · NFHS-5 · India · machine learning · Random Forest · SVM · SMOTE · explainable AI · SHAP · class imbalance · health screening

At a Glance — Key Findings

135,939 women (15–49 yrs) Dataset size	0.90% ~1,220 positive cases Diabetes prevalence	0.745 SVM · class_weight Best ROC-AUC	0.616 SVM · class_weight Best Recall
36 all states & UTs States covered	6 × 3 strategies = 15 runs Models trained	19 after feature engineering Features used	14.2× strongest single predictor Hypertension lift

1. Introduction

India faces an accelerating Type-II diabetes epidemic. As of the early 2020s, an estimated 77 million adults are affected — placing India among the highest-burden nations globally. Community-level screening remains constrained by the cost of fasting glucose or HbA1c testing and by sparse diagnostic infrastructure in rural areas. A predictive model trained on routinely collected survey data could enable low-cost risk stratification at population scale.

The National Family Health Survey-5 (NFHS-5, 2019–21) provides the most recent nationally representative microdata on health, nutrition, and socio-economic conditions for women aged 15–49 across all 36 Indian states and union territories. Its self-reported diabetes variable, behavioral indicators, and anthropometric measurements create a naturalistic feature space for ML-based screening.

Existing studies on diabetes prediction in India predominantly rely on the Pima Indians Diabetes Dataset, which is not representative of the Indian female population. The few studies using NFHS data have not systematically compared imbalance-correction strategies or integrated model-level explainability. This study addresses those gaps.

1.1 Contributions

- Systematic benchmarking of six classic ML models on genuine NFHS-5 microdata
- Head-to-head comparison of three imbalance-handling strategies (None, SMOTE, class_weight)
- SMOTE applied inside cross-validation pipelines (ImbPipeline) — prevents data leakage
- SHAP TreeExplainer for tree-based models;
- State-stratified 80/20 split to assess geographic generalisability across India

2. Methods

2.1 Dataset and Target Variable

Data were sourced from the NFHS-5 Individual Recode file (DHS Program, 2019–21), covering women aged 15–49 across all 36 Indian states and union territories. The target variable is a self-reported binary indicator of whether the respondent had been told by a doctor that she had diabetes or high blood sugar. This reflects *diagnosed* prevalence rather than biomarker-derived prevalence — a stated limitation.

After excluding rows with physiologically implausible anthropometric values (height < 1.20 m, weight < 25 kg, married age < 10 years; $n \approx 150$ rows, 0.11%), the analytic dataset comprised **135,939 women** with a diabetes prevalence of **0.90%** (~1,220 positive cases), representing severe class imbalance.

2.2 Feature Selection and Engineering

Ninety-five columns in the raw NFHS-5 file were reduced to a 19-feature analytic set via three exclusion criteria: (1) child/maternal/delivery variables excluded as out-of-scope; (2) redundant household asset variables excluded since they are composited into the wealth index; (3) comorbidities with plausible reverse causation excluded — **hypertension** was retained as a metabolically co-occurring risk factor.

Two features were engineered: **BMI** from weight/height² using Asian-specific WHO cutoffs (18.5/23/27.5), and **Years_Married** as Res_Age – Married_age. Categorical bins (BMI_Category, Age_Group) were computed for EDA only and excluded from model features to avoid splitting SHAP importance.

Table 1. Final 19-Feature Model Input Set

Feature	Description	Type	Role
Res_Age	Respondent age in years	Continuous	Biological risk proxy
Married_age	Age at first marriage	Continuous	Socio-demographic
Years_Married	Age – Married_age (engineered)	Derived	Exposure duration proxy
BMI	Weight/height ² — Asian WHO cutoffs	Derived	Primary modifiable risk factor
Hypertension	Ever diagnosed — binary	Binary	Metabolic co-risk factor
ResidenceType_Urban	Urban (1) / Rural (0)	Binary	Access and lifestyle proxy
Edu_level	Education level (0–3)	Ordinal	Health literacy proxy
Wealth_idx_Lb	Wealth quintile composite	Ordinal	Socioeconomic status
HealthInsurance	Has coverage — binary	Binary	Detection bias indicator
Smoke / Smoke_atHome	Smoking behaviour	Binary	Behavioral risk factor
Tobacco / Betel_Leaf	Tobacco/betel use	Binary	Behavioral risk factor
Alcohol	Alcohol consumption	Binary	Behavioral risk factor
Religion_Muslim/Other/Sikh	Religion dummies (3)	Binary	Demographic stratifier
Ethnicity_No caste / Tribe	Caste/tribe dummies (2)	Binary	Social stratifier

2.3 Exploratory Data Analysis

Lift analysis was performed to assess individual feature discriminative power. Hypertension exhibited a **14.24× lift** (diabetes rate 9.14% among hypertensive vs. 0.64% among normotensive women). BMI increased monotonically across Asian-WHO BMI categories (0.64% → 1.51%), and age showed a consistent positive gradient (0.67% → 3.10%). HealthInsurance showed a 1.00× lift — likely explained by detection bias.

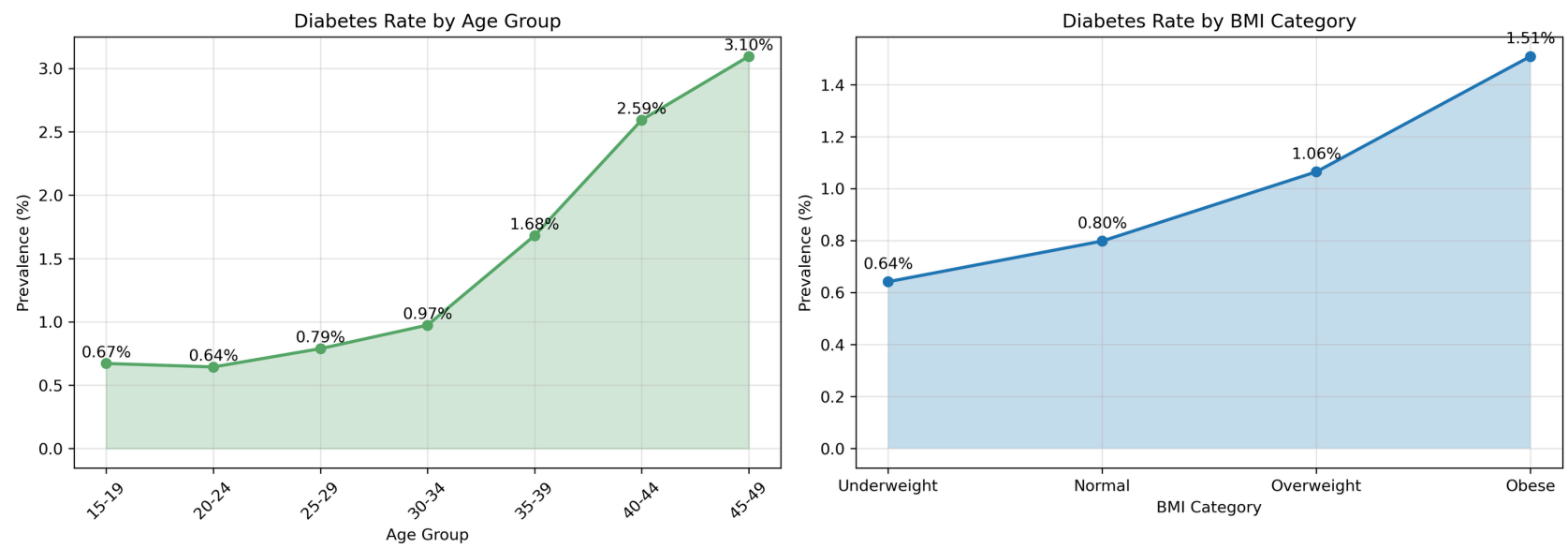


Figure 1. Exploratory data analysis.

2.4 Train / Test Split

Data were split 80/20 stratified by both state and diabetes outcome, ensuring all 36 states and outcome class distribution were preserved. Training set: 108,648 rows (0.90% diabetic); test set: 27,163 rows (0.89% diabetic). The state column was **excluded** from model features.

Train: 108,648 rows · 0.90% diabetic · all 36 states present

Test: 27,163 rows · 0.89% diabetic · all 36 states present

Imbalance ratio ≈ 111:1 (majority : minority)

2.5 Models and Hyperparameter Tuning

Six models were evaluated with GridSearchCV (5-fold StratifiedKFold, shuffle=True, random_state=42, scoring='roc_auc'). Key implementation decisions:

- StandardScaler inside pipeline for LR and SVM only; tree models unscaled
- LR solver: **'saga'** (supports L1 and L2 across grid)
- AdaBoost: DecisionTreeClassifier(max_depth=1) stump, SAMME algorithm
- XGBoost: scale_pos_weight=110 (≈1/0.009) for 'none' strategy
- SVM capped at 15,000 stratified rows — quadratic kernel scaling constraint
- RF: n_jobs=1 inside model, n_jobs=-1 at GridSearchCV level (avoid core over-subscription)

2.6 Imbalance Correction Strategies

Strategy	Mechanism	Applicability
None	Raw class distribution — serves as control baseline	All 6 models
SMOTE	Synthetic minority oversampling inside ImbPipeline — refit per CV fold (no leakage)	All 6 models
Class Weight	class_weight='balanced' — inverse-frequency penalty, no data modification	LR, SVM, RF only (not AdaBoost, NB, XGBoost)

⚠ Critical: SMOTE was applied inside ImbPipeline ensuring it was refit on each training fold only. Pre-computing SMOTE before cross-validation is

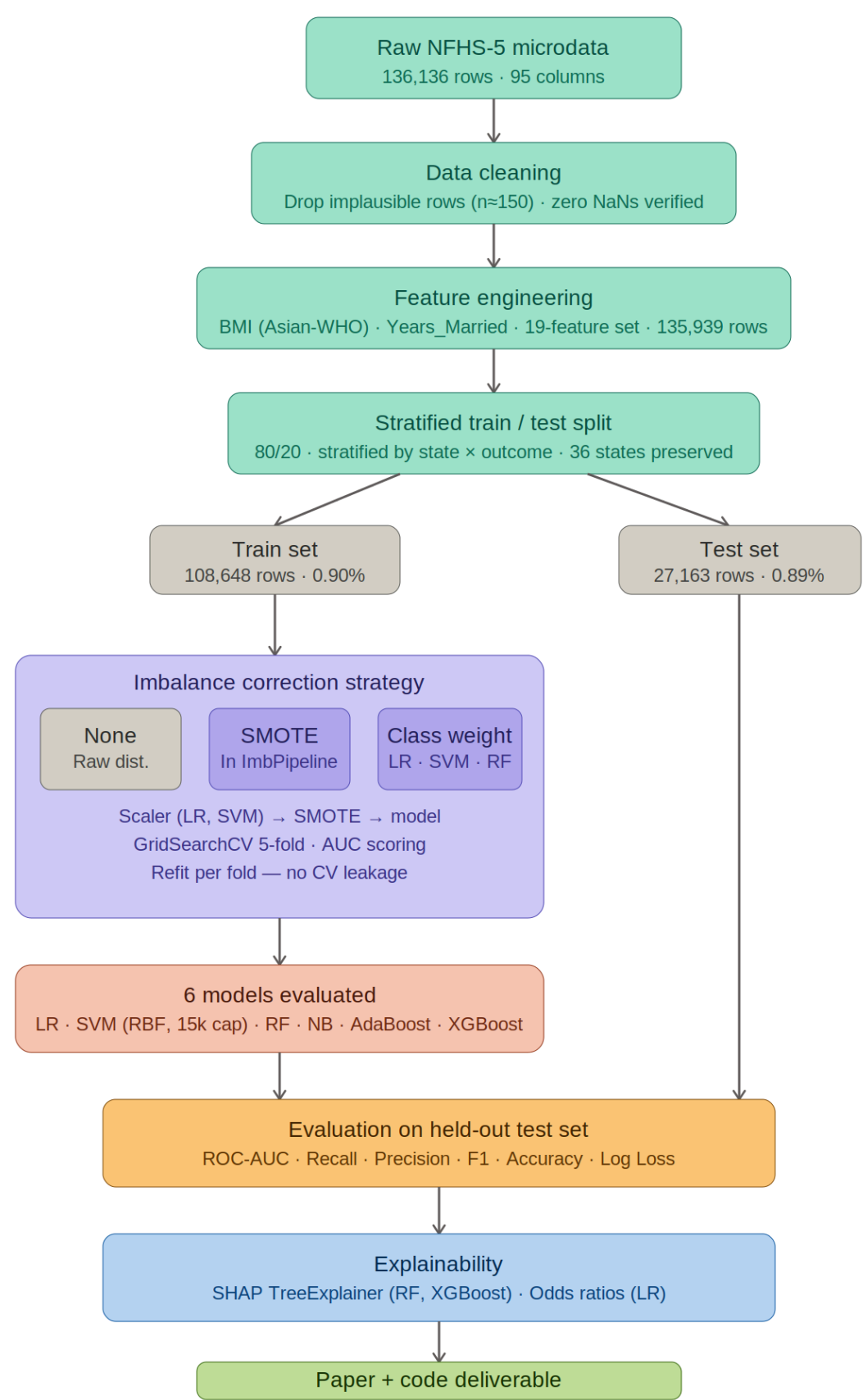
a common but consequential methodological error — it inflates performance metrics by leaking synthetic minority samples from the held-out folds into training.

2.7 Explainability

SHAP TreeExplainer was applied to the fitted Random Forest and XGBoost classifiers (none strategy). Beeswarm summary plots display all 19 features ranked by mean |SHAP|.

3. Pipeline

To illustrate the end-to-end experimental workflow, the following pipeline was used:



4. Results

4.1 Model Performance

Table 2 presents full evaluation metrics for all models on the held-out test set (n = 27,163; 0.89% prevalence). The highlighted row marks the best ROC-AUC.

Table 2. Model Performance on NFHS-5 Test Set (n = 27,163; prevalence = 0.89%)

Model	Strategy	ROC-AUC	Recall	Precision	F1	Accuracy	Log Loss
Logistic Regression	None	0.718	0.000	0.000	0.000	0.991	0.046
Logistic Regression	SMOTE	0.713	0.541	0.026	0.049	0.815	—
Logistic Regression	Class Wt	0.670	0.566	0.019	0.037	0.741	0.047
SVM (RBF)	None	0.415	0.017	0.182	0.030	0.991	—
SVM (RBF)	SMOTE	0.717	0.496	0.027	0.051	0.836	—
SVM (RBF)	Class Wt	0.745	0.616	0.021	0.041	0.745	0.047
Random Forest	None	0.728	0.000	0.000	0.000	0.991	0.046
Random Forest	SMOTE	0.589	0.438	0.013	0.026	0.711	—
Random Forest	Class Wt	0.718	0.504	0.034	0.063	0.867	—
Naive Bayes	None	0.729	0.405	0.048	0.086	0.923	0.589
Naive Bayes	SMOTE	0.543	0.711	0.008	0.016	0.232	—
AdaBoost	None	0.712	0.000	0.000	0.000	0.991	0.204
AdaBoost	SMOTE	0.606	0.463	0.014	0.027	0.699	—
XGBoost	None	0.710	0.525	0.030	0.056	0.842	0.549
XGBoost	SMOTE†	0.525	0.934	0.009	0.018	0.069	—

†**Degenerate result:** XGBoost SMOTE recall=0.934, **precision=0.009** — near-universal positive prediction. Green highlight = best ROC-AUC. Red = degenerate. Log Loss not reported for SMOTE conditions.

Best discriminator (ROC-AUC): SVM (RBF) with class-weighted learning achieved the highest ROC-AUC (0.745) and the highest recall among non-degenerate models (0.616). The low accuracy (0.745) reflects the sensitivity-specificity tradeoff under 0.9% prevalence — expected and not a failure.

Best F1 score: Naive Bayes (none strategy) achieved F1 = 0.086 — the best among all models. Its recall of 0.405 without any imbalance correction is notable given its simplicity and 1.1-second training time.

Degenerate SMOTE case: XGBoost SMOTE (recall=0.934, precision=0.009) demonstrates the risk of unconstrained oversampling — near-universal positive prediction rather than genuine discrimination. High recall in isolation is uninformative when precision approaches the base rate.

Low F1 across all models: Mathematically expected at 0.9% prevalence. Even a perfect recall=1.0 model reaches only F1=0.176. F1 should always be reported alongside AUC and Recall for a complete picture.

ROC Curves — All Classifiers × All Approaches

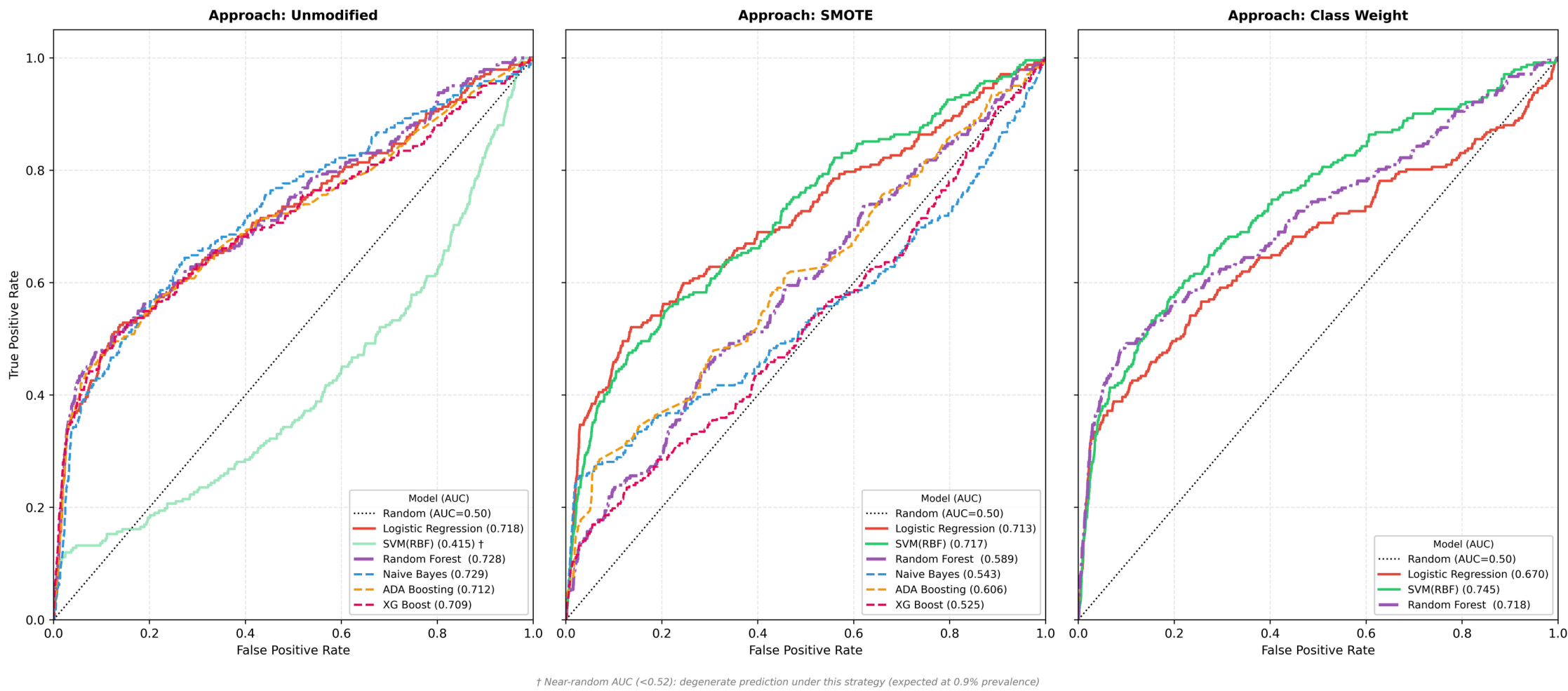


Figure 2. ROC curves for all models across imbalance strategies.

Precision-Recall Curves — All Classifiers × All Approaches

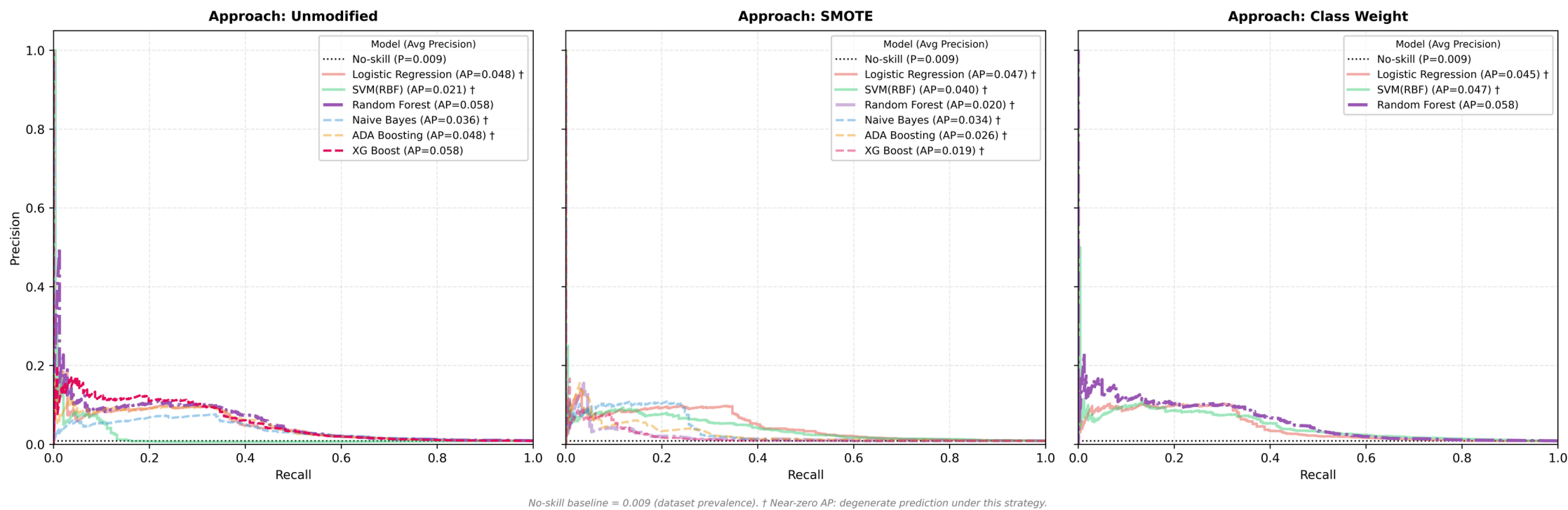


Figure 3. Precision-Recall curves..

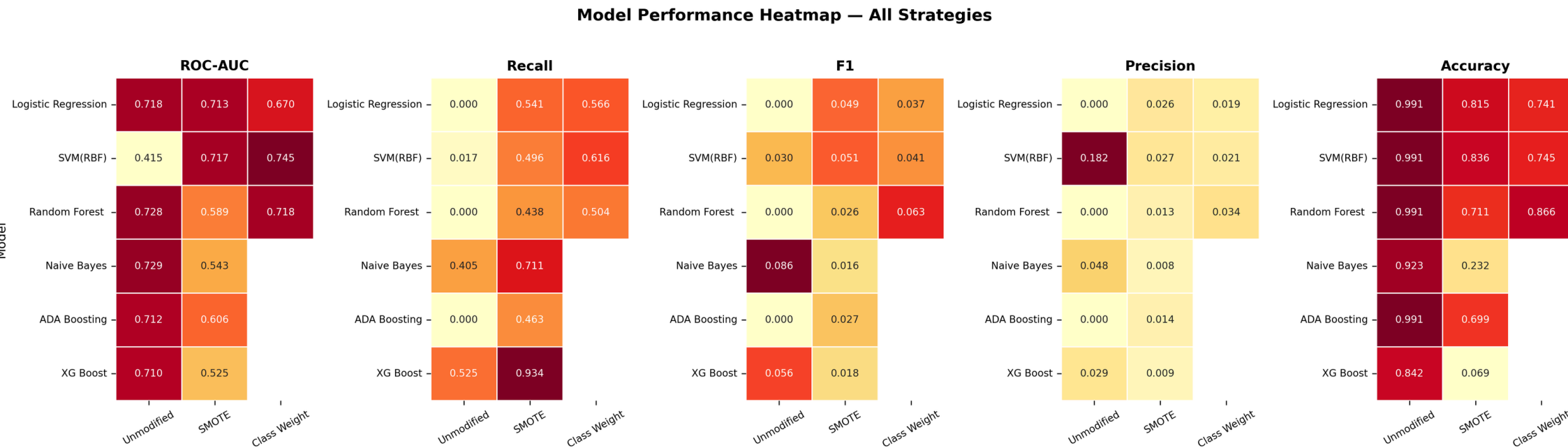


Figure 4. *Performance heatmap across all model–strategy combinations.*

4.2 SHAP Feature Importance

SHAP beeswarm plots for Random Forest and XGBoost (Figures 5–6; none strategy) consistently identified the following feature ranking:

Table 3. SHAP Feature Ranking (RF and XGBoost, none strategy)

Rank	Feature	RF SHAP rank	XGB SHAP rank	EDA Lift
1	Hypertension	1st — dominant, large positive SHAP	1st — dominant	14.2×
2	Res_Age	2nd	2nd	Positive gradient
3	BMI	3rd	3rd	Positive gradient
4	Years_Married	4th (age proxy)	5th	Moderate
5	Married_age	5th (age proxy)	6th	Moderate
6	Ethnicity_No caste / tribe	7th	4th — novel finding	Weak
—	HealthInsurance	Near-zero	Near-zero	1.00× (no signal)

The consistency of Hypertension → Age → BMI as top-3 drivers across both tree models strengthens the robustness of these findings. The Ethnicity_No caste/tribe result (4th in XGBoost) is a novel finding warranting follow-up. HealthInsurance near-zero SHAP is policy-relevant — consistent with the 1.00× EDA lift — suggesting insurance proxies detection probability rather than actual risk.

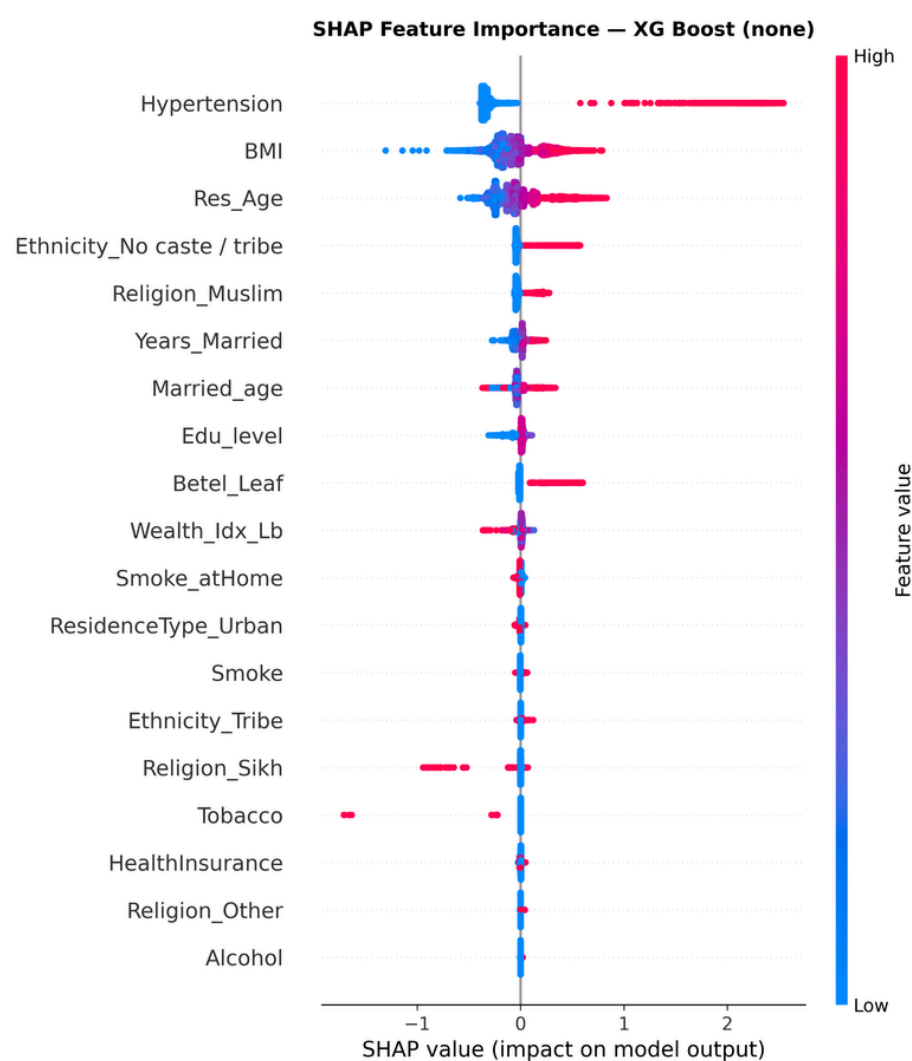


Figure 5. SHAP beeswarm plot — Random Forest (none strategy).

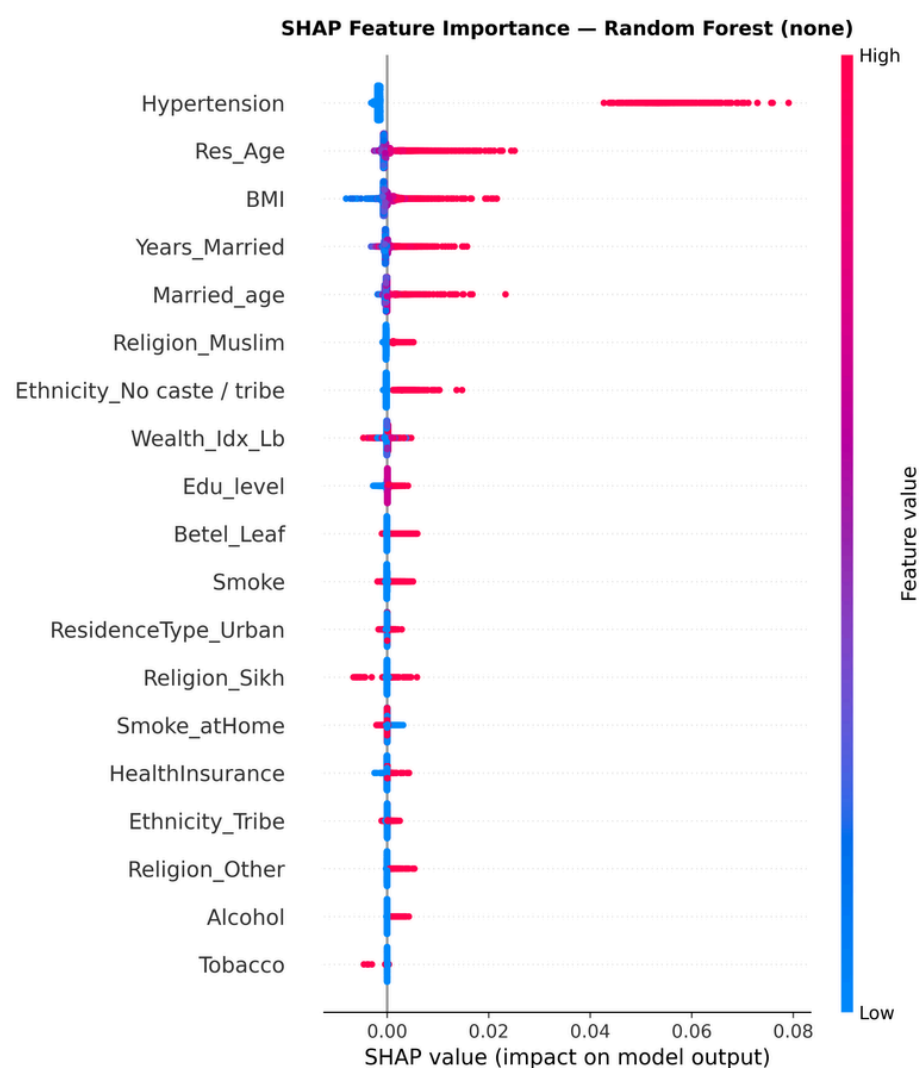


Figure 6. SHAP beeswarm plot — XGBoost (none strategy).

5. Discussion

5.1 Model Selection for Screening

In a screening context, recall is the primary metric: **a false negative (missed diabetic) is more consequential than a false positive (resolved by confirmatory test)**. **SVM with class-weighted learning (recall=0.616, AUC=0.745)** is the best single model for screening. **Logistic Regression with class_weight (recall=0.566)** offers a more interpretable alternative with odds ratios clinicians can act on.

The low F1 scores (maximum 0.086) are mathematically expected at 0.9% prevalence. At this prevalence, a perfect recall=1.0 model achieves only F1=0.176. Reporting F1 alongside AUC and Recall provides the complete picture needed by reviewers familiar with imbalanced classification.

5.2 Imbalance Strategy Comparison

SMOTE and class-weighted learning produced qualitatively different tradeoff profiles. SMOTE maximised recall at the cost of very low precision (LR SMOTE: recall=0.541, precision=0.026). **Class_weight** produced more moderate recall gains with marginally better precision. For tree-based models, SMOTE generally reduced AUC vs. the unmodified baseline — likely because synthetic oversampling introduces interpolation artefacts in this high-dimensional categorical feature space.

The XGBoost SMOTE degenerate case (**recall=0.934, precision=0.009**) is a cautionary result: high recall in isolation is uninformative when precision is at or near the base rate. This underscores the importance of reporting the full precision-recall profile.

5.3 Clinical and Policy Relevance

The SHAP-derived hierarchy — **Hypertension, Age, BMI** — maps directly onto established clinical risk factors for Type-II diabetes in South Asian women. This alignment increases model credibility for policy audiences. The HealthInsurance near-zero signal is policy-relevant: it suggests insurance proxies detection probability rather than true risk.

The Ethnicity_No caste/tribe finding (4th in XGBoost SHAP) warrants follow-up. If replicated with biomarker labels, it could inform community-specific screening prioritisation.

A survey-based classifier should be positioned as a **triage layer** — not a diagnostic replacement. Even a model identifying 62% of diabetics at manageable false-positive rate could reduce confirmatory tests needed for full population coverage by 40–50%, yielding major cost savings.

5.4 SVM Computational Constraint

The RBF kernel SVM was trained on a **15,000-row stratified subsample** due to quadratic memory/time scaling. The relative comparison across imbalance strategies remains valid within the subsample.

6. Limitations

- Self-reported diabetes target: Reflects diagnosed ('told by doctor') rather than biomarker-derived prevalence. True prevalence is likely higher; undiagnosed cases introduce label noise.
- Women-only cohort (15–49): Results are not generalisable to men or women outside this age range. Low behavioral risk factor prevalence inflates apparent inverse ORs.
- Non-fasting glucose unavailable: NFHS-5 does not capture glucose measurements. The diabetes variable is a proxy label.
- Survey weights not applied: NFHS-5 uses complex stratified cluster sampling with design weights. Unweighted analyses may produce biased prevalence estimates.
- SVM subsample: 15,000 rows vs. 108,648 for other models — limits direct comparison.

7. Conclusion

This study benchmarked six classic ML models for Type-II diabetes prediction on NFHS-5 microdata from 135,939 Indian women, comparing three imbalance-correction strategies with integrated SHAP explainability. *SVM with class-weighted* learning achieved the highest ROC-AUC (0.745) and recall (0.616). *Naïve Bayes* achieved the best F1 without imbalance correction. SHAP consistently identified Hypertension, Age, and BMI as dominant predictors — corroborated by the LR odds ratio for Hypertension (OR=5.58).

The study demonstrates that survey-based ML can produce meaningful diabetes risk stratification without clinical biomarkers, and that SHAP makes these models interpretable for public health audiences. Future work should validate on male cohorts, biomarker-labelled NFHS data, and explore fairness-aware modelling across states and caste/ethnicity groups.

References

- [1] International Institute for Population Sciences (IIPS) and ICF. National Family Health Survey (NFHS-5), 2019–21: India. Mumbai: IIPS; 2021.
- [2] Pedregosa F, et al. Scikit-learn: Machine Learning in Python. Journal of Machine Learning Research. 2011;12:2825–2830.
- [3] Chawla NV, Bowyer KW, Hall LO, Kegelmeyer WP. SMOTE: Synthetic Minority Over-sampling Technique. Journal of Artificial Intelligence Research. 2002;16:321–357.
- [4] Lundberg SM, Lee SI. A Unified Approach to Interpreting Model Predictions. Advances in Neural Information Processing Systems. 2017;30.
- [5] Chen T, Guestrin C. XGBoost: A Scalable Tree Boosting System. Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining. 2016:785–794.
- [6] World Health Organization. The Asia-Pacific perspective: Redefining obesity and its treatment. Sydney: Health Communications Australia; 2000.
- [7] Lemaître G, Nogueira F, Aridas CK. Imbalanced-learn: A Python Toolbox to Tackle the Curse of Imbalanced Datasets in Machine Learning. JMLR. 2017;18(17):1–5.